a Four streams per view: photograph, depth frame, point cloud and the recipe as text (a plant line, then one line per leaf; maize: 5 of its 7 leaf fields)

sorghum

photograph



depth



point cloud



recipe

```
stem 1.34 m, 24 leaves
leaf start  len roll branch width
   1  0.20 0.24   65     20 0.080
   2  0.22 0.36  186     21 0.088
   3  0.24 0.46   41     21 0.094
   4  0.27 0.60  228     21 0.094
   5  0.30 0.62   36     21 0.113
   6  0.33 0.67  185     21 0.094
     … 18 more leaves
```

maize

photograph



depth



point cloud



recipe

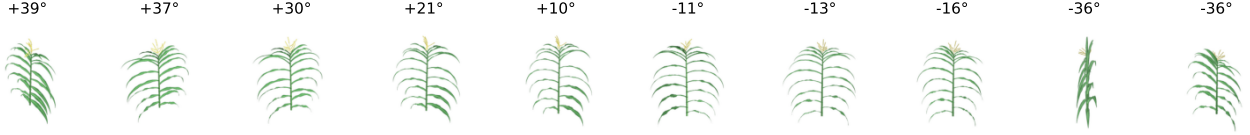
```
height 0.95 m, 18 leaves
leaf  dist  len   wid angle droop
   1 0.059 0.33 0.043    -1   -84
   2 0.059 0.43 0.056    -8   -87
   3 0.059 0.44 0.038     4   -98
   4 0.059 0.39 0.051     5   -94
   5 0.059 0.45 0.049     2   -91
   6 0.059 0.47 0.040    10   -91
     … 12 more leaves
```

b Ten cameras per plant, labelled by camera elevation above the plant: sorghum on a fixed ladder, maize at random elevations and azimuths (sorted here)

sorghum



maize



c Rows: the target between two neighbours. Red = neighbour pixels, orange = target pixels hidden behind a neighbour, grey points = neighbours

sorghum



maize

